

Exam in Statistical Physics and Solid State Physics

Exam date: xx.1.2025

Exam time: 4.5 hours.

Permitted help: books, notes, computer with ITX Flex running, camera for scanning the results. No phones, no internet apart from what is needed for the digital exam.

Answers can be given in English or Danish.

1 Crystal Structure (4 points)

Figure 1 shows a two-dimensional crystal. All the atoms shown are of the same chemical element.

1. Add the lattice vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ spanning the Bravais lattice to Fig. 1. **1 p: Horizontal length a , vertical length b .**
2. How many atoms does the basis contain? **1 p: 2**
3. In an additional figure, draw the corresponding reciprocal lattice vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ and give expressions for their length. **1 p: Horizontal length $2\pi/a$, vertical length $2\pi/b$.**
4. Now assume that the displacement d is taken to $d = 0$. Draw the reciprocal lattice vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ and give expressions for their length in this case. **1 p: Horizontal length $4\pi/a$, vertical length $2\pi/b$.**

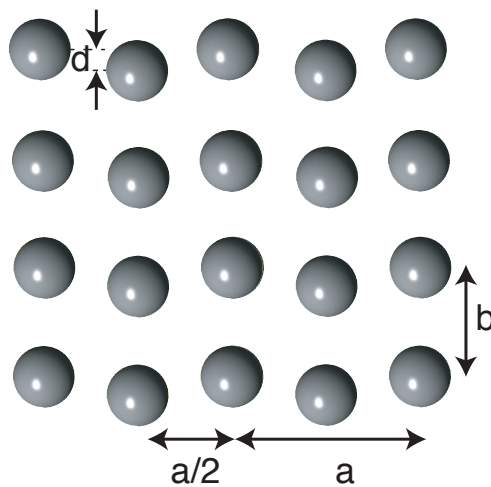


Figure 1: Two-dimensional crystal.

2 The reciprocal lattice (2 points)

Consider a three-dimensional Bravais lattice $\mathbf{R} = m\mathbf{a}_1 + n\mathbf{a}_2 + o\mathbf{a}_3$ and the corresponding reciprocal lattice $\mathbf{G} = m'\mathbf{b}_1 + n'\mathbf{b}_2 + o'\mathbf{b}_3$. Using the explicit definition of the reciprocal lattice vectors

$$\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}, \quad \mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}, \quad \mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}, \quad (1)$$

- Show that $\mathbf{b}_2$ is perpendicular to $\mathbf{a}_3$. 0.5 p: because of cross product
- Show that $\mathbf{b}_2$ is perpendicular to $\mathbf{a}_1$. 0.5 p: because of cross product
- Does this imply that $\mathbf{b}_2$ is parallel to $\mathbf{a}_2$? Give a brief reason. 1 p: No. $\mathbf{b}_2$ is only parallel to $\mathbf{a}_2$ if $\mathbf{a}_2$ is also perpendicular on $\mathbf{a}_1$ and $\mathbf{a}_3$, like in a cubic lattice. But this doesn't have to be the case.

3 Vibrations (2 points)

Figure 2 shows the vibrational modes for a one-dimensional chain with two atoms per unit cell. The unit cell length is b .

- The two atoms in the unit cell have different masses M_1 and M_2 . Draw the dispersion in case that $M_1 = M_2$. 1 p. The gap would vanish or one goes one step further, argues that the unit cell is now $b/2$ with one atom per unit cell, unfolds it and has a single acoustic branch with a BZ size of $4\pi/b$.
- Suppose that you use periodic boundary conditions and that you have N unit cells in the one-dimensional solids. How many different normal modes do you get? 1 p. $2N$. Two per cell (one for each atom) and then as many as I have cells.

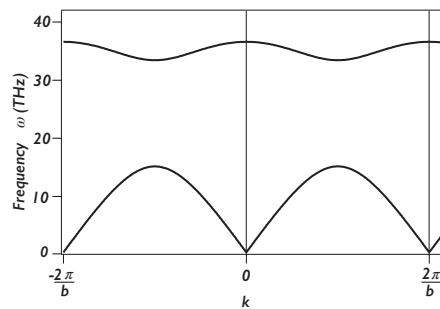


Figure 2: Vibrational modes for a one-dimensional solid.

4 Heat capacity (2 points)

Consider the temperature-dependent heat capacity of three three-dimensional solids in Fig. 3. Assume that the highest temperature (10 K) is much lower than the Debye temperatures of these solids.

- Solid 1 is an insulator and solids 2 and 3 are metals. Explain how this can be deduced from the figure. **1 p: From the slope of the curves in the log-log plot. The T -dependence of solids 2 and 3 is $C \propto T$ whereas it is $C \propto T^3$ for solid 1.**
- In what way are the two metals (solid 2 and solid 3) different from each other? **1 p: Solid 2 has a higher density of states at the Fermi energy.**

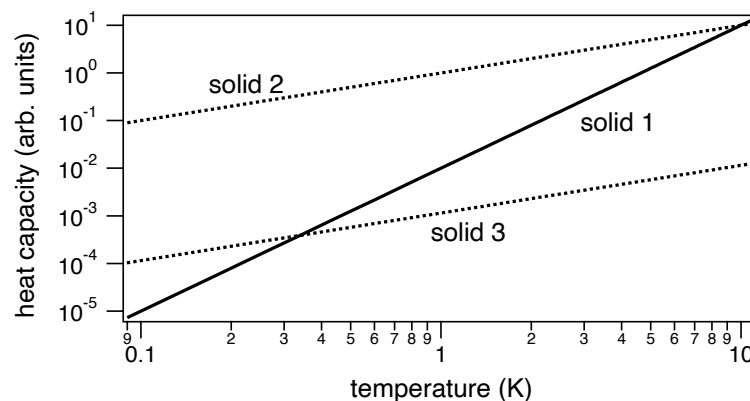


Figure 3: Temperature-dependent heat capacity of three solids.

5 Conductivity (1.5 p)

Consider a metallic wire of length l and diameter d . The resistance of the wire is:

$$R = \frac{4lm_e}{\pi d^2 e^2 n \tau}, \quad (2)$$

where e is the charge of the particles conducting the current, m_e is their mass, n is their density, and τ is the relaxation time.

- Give an expression for the wire's conductivity. **0.5 p, no point if conductivity is given a inverse of resistance.**

$$\sigma = \frac{e^2 n \tau}{m_e}, \quad (3)$$

- How does the conductivity depend on the temperature in the Drude model? For a correct answer, it is sufficient to give the functional dependence without any constants, e.g. it depends on the temperature as $\propto T^2$ or $\propto T$ or not at all. Give a reason for your answer. **1 p. The only T-dependent part is τ and $\sigma \propto \tau$. But $\tau \propto 1/v_t$ and $v_t \propto \sqrt{T}$. So as a result $\sigma \propto 1/\sqrt{T}$.**

6 Bloch theorem (2 points)

The general form of an electronic wave function in a crystalline solid is $\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}u_{\mathbf{k}}(\mathbf{r})$, where $u_{\mathbf{k}}(\mathbf{r})$ is a lattice-periodic function.

- Let $\mathbf{R}$ be a vector of the crystal's Bravais lattice. Show that the Bloch wave function calculated at the point $\mathbf{r} + \mathbf{R}$ only differs by a factor f from the Bloch wave calculated at $\mathbf{r}$, i.e., $\psi_{\mathbf{k}}(\mathbf{r} + \mathbf{R}) = f\psi_{\mathbf{k}}(\mathbf{r})$. **1 p: $\psi_{\mathbf{k}}(\mathbf{r} + \mathbf{R}) = e^{i\mathbf{k}\cdot(\mathbf{r}+\mathbf{R})}u_{\mathbf{k}}(\mathbf{r} + \mathbf{R}) = e^{i\mathbf{k}\cdot(\mathbf{r}+\mathbf{R})}u_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{R}}\psi_{\mathbf{k}}(\mathbf{r})$**
- What is the magnitude of this factor f ? **1 p: 1. It's just a complex phase factor.**

7 Band structure (3.5 points)

Figure 4 shows the band structure of the semiconductor ZnSe along different directions in the Brillouin zone. The energy zero is the top of the valence band.

1. How many valence electrons are there per unit cell (at least)? **1 point: There are four occupied bands (at least) with 2 electrons per unit cell per band, so at least 8 electrons per unit cell.**
2. Is the band gap is direct or indirect? **0.5 point: direct**
3. What is the group velocity of the electrons and the holes at the conduction band minimum and valence band maximum, respectively? **1 point: both zero because the bands have a minimum and a maximum, respectively.**
4. What is the relation between their effective masses, i.e. is the electron or hole effective mass higher in this material (again, at the conduction band minimum and valence band maximum)? Give a brief reason. **1 point: The hole effective mass is higher. The effective mass is proportional to the inverse curvature of the bands. The hole band / valence band is flatter, its curvature is lower and hence its inverse curvature is higher.**

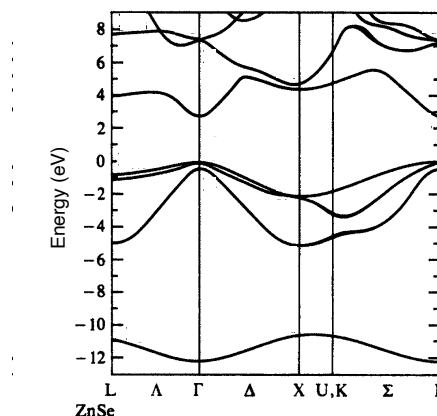


Figure 4: Band structure of ZnSe.

8 Superconductivity (1 point)

The transition from the normal to the superconducting state is associated with an energy lowering. By how much is the energy of an electron lowered when it forms a Cooper pair with another electron? Give the order of magnitude in electron volts and explain briefly how you arrive at that number.

The order of magnitude is Δ , (half) the size of the superconducting gap. This is on the order of a few meV or less. I can either arrive at this by arguing that the gap must be smaller than energy corresponding to a typical T_c or by saying that it must be a phonon energy scale because the phonons mediate the superconductivity.)

9 Partition function

1) The possible energies are -2ϵ , 0 , 0 , and $2\epsilon + \Delta$, so the partition function is

$$Z = e^{2\beta\epsilon} + 2 + e^{-\beta(2\epsilon+\Delta)}, \quad \beta = \frac{1}{kT}. \quad (1)$$

2)

$$\bar{E} = \frac{-2\epsilon e^{2\beta\epsilon} + (2\epsilon + \Delta)e^{-\beta(2\epsilon+\Delta)}}{e^{2\beta\epsilon} + 2 + e^{-\beta(2\epsilon+\Delta)}} \quad (2)$$

$$\lim_{T \rightarrow 0} \bar{E} = -2\epsilon, \quad \lim_{T \rightarrow \infty} \bar{E} = \frac{\Delta}{4} \quad (3)$$

For $T \rightarrow 0$, the system approaches the ground state. For $T \rightarrow \infty$, all states are equally likely, as this maximizes the entropy.

3)

$$S = \frac{\bar{E}}{T} + k \ln(Z) = \frac{-2\epsilon e^{2\beta\epsilon} + (2\epsilon + \Delta)e^{-\beta(2\epsilon+\Delta)}}{T(e^{2\beta\epsilon} + 2 + e^{-\beta(2\epsilon+\Delta)})} + k \ln(e^{2\beta\epsilon} + 2 + e^{-\beta(2\epsilon+\Delta)}) \quad (4)$$

10 Two-state paramagnets in thermal equilibrium

1)

$$S = 2k \ln \left(\frac{N!}{(N - N_0)!N_0!} \right) \quad (5)$$

2) Initially, the total energy of the system is

$$E = -\mu B(2N_0 - N) - \mu_x B(2N_0 - N). \quad (6)$$

At the later time, the energy is

$$E = -\mu B(2N_1 - N) - \mu_x B(2N_2 - N). \quad (7)$$

Due to energy conservation, the energy must be the same. Equating the two expressions and rearranging gives

$$N_1 = (1 + x)N_0 - xN_2. \quad (8)$$

3)

$$\left(\frac{N - N_1}{N_1} \right)^x = \frac{N - N_2}{N_2} \quad (9)$$

11 Photons

1)

$$N = 2.40 \cdot 8\pi V \left(\frac{kT}{hc} \right)^3, \quad (10)$$

where k is Boltzmann's constant, h is Planck's constant, and c is the speed of light in vacuum.